

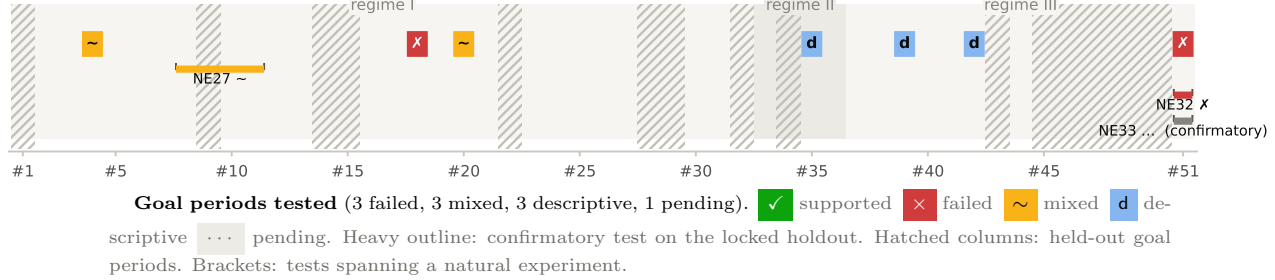
H122: A batch join is a spin addition: do incumbents respond as the fitted couplings say? (NE27, NE33)

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1.5/5 Usefulness 1/5

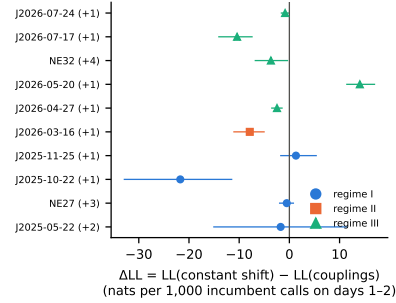
Bottom line. Fitted newcomer couplings do not predict incumbents' first two days after a join better than a constant field shift (6/10 events favour the shift; NE32 $\Delta LL = -3.7$ [-6.9, -0.2]); the HH fails.

level: hypothesis · pipeline checks 84% · updated 2026-10-04



WHAT HAPPENED

When agents join, a kinetic Ising model adds J_{iNs_N} to every incumbent's field. HH363 predicts that newcomer couplings fitted on days 3–7 after a join, applied back to days 1–2, predict the incumbents' per-call talk better than a constant field shift. The test uses 10 non-holdout join events (4 in regime I, 1 in II, 5 in III), with a log-loss cross-fit on held-out days. **The HH fails.** The constant shift wins in 6 of 10 events, including NE32, the stand-in for NE33 ($\Delta LL = -3.7$ [-6.9, -0.2] nats per 1,000 calls; the kill fires). Couplings win in 1. Without the #51 kickoff day in NE32's baseline, the result weakens to -1.6 [-3.8, +0.7] (post hoc). NE33 itself is held out and blind.



Observables. Each row is one join event (number of newcomers in brackets), with ΔLL and its 95% block-bootstrap CI. Values left of zero mean the constant shift predicts days 1–2 better than the fitted couplings. Six events sit left with CIs excluding zero, one sits right (Gemini 3.5 Flash, 05-20), and three include zero (all small regime-I joins).

PHYSICAL MODEL

Kinetic Ising on the call clock (model 02). Incumbent i 's receiving call c has talk spin Y_{ic} . Its field sums incumbent baselines $a_{i,\kappa}$ and the messages read at the call (R , ledger rule). The H18 dilution factor scales read couplings. Newcomers add a read coupling J_N (MJ) or a uniform shift δ (MC). The in-flight placebo P^N counts newcomer messages posted during the call: unreadable, at matched lag.

$$\text{logit } P(Y_{ic}=1) = a_{i,\kappa} + \left(\frac{\bar{N}_{post}}{\bar{N}_{pre}} \right)^{-0.66} (\beta_m R_{ic}^m + \beta_o R_{ic}^o + \beta_P P_{ic}) + \gamma \cdot E_{ic} + \rho Y_{i,c-1} + \phi(k)$$

$$\begin{aligned} \text{MC: } & + \delta & \text{MJ: } & + J_N R_{ic}^N & \text{MJ}_{pl}: & + J_P P_{ic}^N \\ \Delta LL = & \langle \ell_{MC} \rangle_{d1-2} - \langle \ell_{MJ} \rangle_{d1-2} & (\delta, J \text{ fit on days 3–7}) \end{aligned}$$

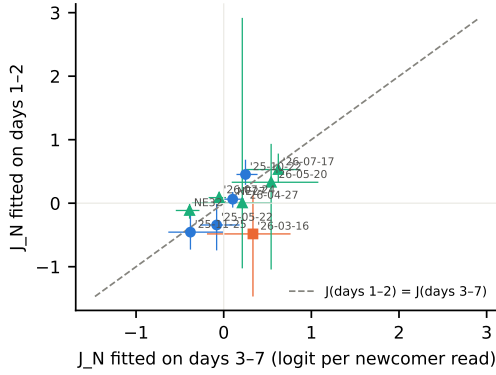
SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator adding agents: the incumbents' first-day response is a uniform step whose sign depends on what else happens that week. A kickoff, a goal change or the end of a kickoff transient matters more than whom the newcomers talk to. Per newcomer message read, an incumbent's talk probability moves by about 0.009 (regime III median). A message that names the incumbent moves it more (named exceeds unnamed in 8/10 events). Practical rule: do not forecast a join's effect from couplings. If you want incumbents to engage a newcomer, have the newcomer name them. Do not schedule a join on a kickoff day if you want to measure it.

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RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
S1–S4 synthetic (NE32)	power 1.00, kill power 1.00, size 0.04; NE27 power 0.47–0.64	pass (NE32)
P1 NE32 MJ beats MC (kill)	−3.68 [−6.92, −0.25]; post hoc w/o kickoff −1.6 [−3.8, 0.7]	failed
P2 NE27: $J_N \approx 0$, step = kickoff	J_N 0.10 [−0.03, 0.23]; step inside 12-kickoff range	supported
P3 fast step	1.82, 0.85, 0.29 (3 resolved)	mixed
$J(1-2)/J(3-7) \in [0.5, 2]$		
P4 read beats in-flight (where MJ wins)	+0.03 [−0.13, 0.20] (1 event)	weak
P5 dilution helps	4/10 events	failed
P6 newcomers couple like incumbents	NE32 −0.36 [−1.02, 0.32]	supported
P7 MJ wins in $\geq 1/2$ of MJ regime III	1/5, MC 4/5	failed
NE33 (confirmation only)	(confirmation not run; frozen confirm.py only)	pending



Second figure. Newcomer read coupling fitted on days 3–7 against the same coupling fitted on days 1–2, per event (95% CIs). Points near the dashed diagonal couple the same from day 1. Couplings exist in some events but are small and change sign across events, so they cannot carry the day-1–2 response.

GOAL PERIODS IN DETAIL

NE33 ... pending (**holdout**)

G04	~ mixed
G18	× failed
G20	~ mixed
G35	d descriptive
G39	d descriptive
G42	d descriptive
G51	× failed
NE27	~ mixed
NE32	× failed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (**writeup/paper.tex**).

CAVEATS

Kickoff fields sit inside several windows (NE27 day 1, two single joins on day 1, NE32's baseline). The constant-shift model absorbs them, so this design cannot separate a join field from a kickoff field. Single joins give 22–447 day-1–2 calls with a newcomer read. NE27 is underpowered and its kill is anti-conservative (size 0.12). The logistic model is linear in clipped read counts. The dilution exponent is H18's, not refitted.

ROUND 2

Run join-free pseudo-joins in regime III to size the calendar drift of δ . Rebuild baselines without kickoff days. Fit a saturating read response. Run NE33 on the holdout with the frozen script.

- **H122-R1** Join-free placebo in regime III.
- **H122-R2** Kickoff-free PRE windows.
- **H122-R3** Saturating read response.
- **H122-R4** NE33 on the holdout with the frozen confirm.py.

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